CODE:

```
import math


board = [" "] * 9

wins = [(0,1,2),(3,4,5),(6,7,8),(0,3,6),(1,4,7),(2,5,8),(0,4,8),(2,4,6)]


def show(b): print("\n".join("|".join(b[i:i+3]) for i in range(0,9,3)))


def win(b):
  for a,b1,c in wins:
    if b[a]==b[b1]==b[c]!=" ": return b[a]
  return None


def moves(b): return [i for i,v in enumerate(b) if v==" "]


def minimax(b, a, beta, maxp):
  if win(b)=="X": return 1
  if win(b)=="O": return -1
  if " " not in b: return 0
  if maxp:
    val=-math.inf
    for m in moves(b):
      b[m]="X"; val=max(val,minimax(b,a,beta,False)); b[m]=" "
      a=max(a,val)
      if beta<=a: break
    return val
  else:
    val=math.inf
```

```
        for m in moves(b):
            b[m]="O"; val=min(val,minimax(b,a,beta,True)); b[m]=" "
            beta=min(beta,val)
            if beta<=a: break
        return val

def best(b):
    sc,m=-math.inf,None
    for i in moves(b):
        b[i]="X"; s=minimax(b,-math.inf,math.inf,False); b[i]=" "
        if s>sc: sc,m=s,i
    return m

def play():
    print("You=O, AI=X"); show(board)
    while not win(board) and " " in board:
        h=int(input("Move 0-8: "))
        if board[h]!=" ": continue
        board[h]="O"
        if win(board) or " " not in board: break
        board[best(board)]="X"; show(board)
    show(board); print("Winner:",win(board) or "Draw")

play()
```

OUTPUT:

```
========== RESTART: C:/Users/arunp/OneDrive/Documents/AI/EXP 14.py ========
You=O, AI=X
 | |
 | |
 | |
Move 0-8: 2
 | |O
 |X|
 | |
Move 0-8: 4
Move 0-8: 5
 | |O
 |X|O
 | |X
Move 0-8: 7
X| |O
 |X|O
 |O|X
X| |O
 |X|O
 |O|X
Winner: X
```